SAMALA SUDHAKAR

AI Engineer | Generative AI | Agentic AI | RAG | Machine Learning | MLOPs | LLMOPs | AI Researcher

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PROFESSIONAL SUMMARY

Machine Learning , AI Engineer and AI Researcher with 7 years of professional experience in scientific data analysis, Python-based data pipelines, automation, statistical modeling, and analytical application development. Hands-on experience building Generative AI, Agentic AI, and Retrieval-Augmented Generation (RAG) applications using LangChain, LangGraph, LangSmith, FAISS, ChromaDB, Hugging Face Embeddings, FastAPI, Streamlit, Docker, MLflow, GitHub Actions, and AWS. Experienced in developing end-to-end ML/AI solutions, deploying APIs, building MLOps workflows, and processing large scientific datasets. Currently pursuing a PhD in Artificial Intelligence.

TECHNICAL SKILLS

Programming & Data: Python, SQL, MySQL, SQLite, Pandas, NumPy, PySpark, ETL, Data Ingestion, Data Preprocessing

Machine Learning: Regression, Classification, Clustering, Time Series Forecasting, Feature Engineering, Model Evaluation, Scikit-learn, XGBoost

Deep Learning & NLP: PyTorch, TensorFlow, CNN, RNN, LSTM, GRU, Transformers, BERT, Tokenization, NLP

Generative AI & LLM: LLMs, GPT, LLaMA, Prompt Engineering, LLM Fine-Tuning, LoRA, PEFT, Embeddings, RAG

Agentic AI & LLM Frameworks: LangChain, LangGraph, LangSmith, Agentic Workflows, Tool Calling, RAG as a Tool, Hugging Face Embeddings

Vector Databases & Search: FAISS, Pinecone, ChromaDB, Vector Search, Semantic Search, Similarity Search

MLOps & Deployment: MLflow, DVC, FastAPI, Docker, CI/CD, GitHub Actions, AWS EC2, AWS S3, AWS EKS

AI Application Development: Streamlit, Pydantic, REST APIs, HTML, CSS, JavaScript, React.js

Monitoring: Prometheus, Grafana

PROFESSIONAL EXPERIENCE

Research Assistant — Central Water and Power Research Station (CWPRS) | 30-09-2019 to Present

- Developed scalable Python data pipelines for processing and analyzing 10,000+ seismic records using Python, Pandas, and MySQL; applied Scikit-learn Linear Regression and least-squares curve fitting to estimate Gutenberg–Richter seismicity parameters.
- Automated data ingestion, preprocessing, statistical analysis, and parameter-estimation workflows, reducing manual effort by 40%.
- Performed statistical modeling and data analysis on earthquake magnitude–frequency datasets to support seismicity assessment and engineering studies.
- Developed Python-based analytical tools and workflows for large scientific datasets and data-driven engineering applications.

Scientific Assistant — Nuclear Power Corporation of India Ltd. | 01-11-2006 to 15-09-2019

- Conducted statistical analysis of large operational datasets for radiation-safety monitoring.
- Developed Python automation scripts for analytics and reporting, improving workflow efficiency.
- Applied computational and data-analysis techniques to scientific and operational datasets.

GENERATIVE AI / MACHINE LEARNING PROJECTS

Corrective Agentic RAG (CRAG) | Generative AI | Agentic AI | RAG | LangChain | LangGraph | MLOps | AWS

- Built an end-to-end Corrective Agentic RAG system for retrieval-augmented question answering using Python, LangChain, LangGraph, FAISS, Hugging Face Embeddings, and Groq LLMs.
- Implemented RAG as a reusable tool within a LangGraph workflow, enabling retrieval-grounded responses and agentic orchestration.
- Developed FastAPI backend APIs and a Streamlit frontend for document upload, conversational interaction, and RAG-based question answering.
- Integrated LangSmith for LLM application tracing and observability.
- Containerized the application with Docker and implemented CI/CD automation using GitHub Actions, with AWS deployment infrastructure.
- Implemented vector-store based document retrieval and worked with embeddings, chunking, and retrieval configuration for RAG.
- Evaluated RAG retrieval across 200+ test queries using Top-K semantic search and LangSmith tracing.

House Price Prediction | Machine Learning | Regression | FastAPI | Docker | AWS

- Built an end-to-end Linear Regression pipeline for house-price prediction, achieving $R^2 = 0.726$ on the housing dataset.
- Developed a FastAPI prediction API, containerized the application using Docker, and deployed it on AWS EC2.

AI Financial Assistant | Deep Learning | Time Series | Sentiment Analysis

- Developed an AI-based financial assistant incorporating deep-learning models for stock-price prediction and integrated sentiment analysis into the workflow.

Airbnb Dynamic Pricing Engine | Machine Learning | Forecasting | FastAPI | Docker

- Developed a machine-learning forecasting solution for dynamic pricing and deployed the application using FastAPI and Docker.

Reservoir-Induced Seismicity Application | Python | Streamlit | FastAPI | Pydantic

- Developed a web-based analytical application for Reservoir-Induced Seismicity using Python, Streamlit, FastAPI, and Pydantic.

EDUCATION

PhD in Artificial Intelligence — Bennett University | 01-08-2025 - Pursuing

PG Diploma in Artificial Intelligence — CDAC Noida | 01-08-2024 to 01-03-2025

Master of Computer Applications (MCA) — Uttaranchal University | 01-08-2023 to 01-09-2025

MSc Physics — Annamalai University | 01-06-2025 to 01-06-2017

CERTIFICATIONS & TRAINING

- Google Advanced Data Analytics — Coursera
- NPTEL: Computer Vision, Natural Language Processing, Generative AI, Big Data Computing, DSA using Python, LLMs, Quantum Computing — 20 Total Credits
- Radiation Protection Training — Bhabha Atomic Research Centre (BARC), Mumbai

INTERNSHIP

- Elevate Labs — Data Analytics | Best Performer
- Unified Mentor — Machine Learning